LOT Pager — Roadmap & Analysis

Document: LOT-HW-01 / ROADMAP · **Rev:** A (2026-11-13)

1. Phases

Phase	Weeks	Output	Gate to pass	Est. spend
S0 Concept (this session)	0	Plan, BOM, specs, PDFs	Decisions D-1…D-4 made	$0
S1 Breadboard	1–3	Stage 0 kit wired; device-sim + `/api/device/*` on staging; Copy → Log tab works end to end	Loop works; Qi-through-steel and RF tests measured	~$170 + parts shipping
S2 EVT (5 units)	4–10	Schematic, 4-layer HDI layout, PCBWay PCBA ×5, CNC enclosure ×5 at 6.0 mm	All functions work in the steel body; battery ≥ 7 days	~$2,500
S3 DVT (15 units)	11–17	Rev B PCB, antenna tuned, 5.0 mm attempt, glass + print, firmware feature-complete, OTA	Drop, IP54, thermal, 2-week field test with 10 users	~$3,500
S4 Certification	14–22 (overlaps)	FCC Part 15 + CE/RED (or modular path), battery UN38.3 report from cell vendor	Test reports + FCC ID	12–20*k* (*or* 3 k modular)
S5 PVT / Run of 100	20–26	PCBWay turnkey 100 + CNC 100 + assembly + test jig	Yield ≥ 95 %, all units pass self-test	~$14,800
S6 Ship	26–28	Boxed units, PDF manuals, Settings → Devices live		—

Total calendar: **~6–7 months** from today to 100 shipped units (mid-2027), if decisions are made in week 1.

2. Decisions needed from the Inventor

ID	Decision	Options	Recommendation
D-1	Where does wireless charging go?	A: face-down through glass · B: ceramic disc in back · C: pogo cradle	**A** — keeps the back pure polished steel
D-2	Thickness	6.0 mm (80 mAh, ~3 weeks) · 5.0 mm (45 mAh, ~2 weeks, custom cell)	Build EVT at 6.0, decide 5.0 after EVT
D-3	Radio certification path	Own FCC/CE cert with PICO-1 SiP (thin) · pre-certified MINI-1 module (thicker, cheaper cert)	PICO-1 if 5 mm matters; MINI-1 if budget matters
D-4	Camera purpose	Photo-to-Log only · on-device AI (e.g. "is it daylight / is someone here")	Photo-to-Log for v1; AI vision as OTA later

3. Risk register

#	Risk	L	I	Mitigation
R1	Steel blocks Qi / FOD stops charging	H	H	D-1 option A; test in S1
R2	Steel detunes 2.4 GHz antenna, poor range	H	M	Antenna corner under glass, isolated bezel, VNA tuning in S3
R3	5 mm not reachable	M	M	Ship at 6.0 mm if needed — still very thin

#	Risk	L	I	Mitigation
R4	Thin LiPo MOQ (often 500–1000)	M	M	Ask PCBWay/cell vendor for 200-unit sample run; pay MOQ premium
R5	BME688 reads self-heat, not room	H	L	Vent + isolated chamber + BSEC compensation; label as "near-device"
R6	Certification cost exceeds unit budget	M	H	D-3; also check FCC rules for a limited first batch with a lab
R7	Mirror polish scratches during assembly	M	L	Protective film from CNC until boxing
R8	Part shortages (Sharp LCD, HM0360)	M	M	Alternatives in BOM §E; buy 120 sets up front
R9	Notice spam annoys users	M	M	Max 6/day, quiet hours, "less often" setting

4. Why this design (short analysis)

- **ESP32-S3** is the cheapest chip with Wi-Fi + BLE + camera input + enough RAM for TLS and JPEG; huge community; PCBWay assembles it daily.
- **Memory LCD** is the only screen that is silver-looking, readable in sunlight, and costs almost no power to keep a message visible all day — ideal for a pager.
- **Bosch BME688 + BHI260AP** are the "AI-grade" off-the-shelf sensors: both run learned models on the sensor itself, so the main chip can sleep.
- **Polling, not push**: predictable battery life; push can come later over the same API.
- **Single vendor (PCBWay)** for PCB, assembly, CNC and polish reduces shipping and coordination for a 100-unit run.